

Quantum Double Models (QDM)

Kitaev, Annals of Phys., 303, 2 (2003)

arXiv 1997

QDM: generalization of $\mathbb{Z}_2$ toric code
to lattice G gauge theory.

→ finite (may be non-Abelian) group.

$$D(G) = G \times \hat{G} \text{ for Abelian } G.$$

$\downarrow$ flux $\downarrow$ charge
(double)

2.1. The model based on a group algebra.

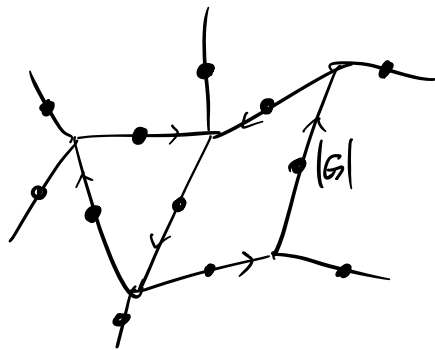
G .

$\mathcal{H} := \mathbb{C}[G] = \left\{ \sum_g a_g \cdot g \mid g \in G, a_g \in \mathbb{C} \right\}$ be the group algebra.
 $\downarrow$
 Hilbert space with orthonormal basis $|g\rangle$
 $\dim \mathcal{H} = |G|$

$$\text{Def: } \begin{aligned} L_+^g |z\rangle &= |gz\rangle & T_+^h |z\rangle &= \delta_{h,z} |z\rangle \\ L_-^g |z\rangle &= |zg^{-1}\rangle & T_-^h |z\rangle &= \delta_{h^{-1},z} |z\rangle \end{aligned}$$

$$\text{Satisfy: } \begin{aligned} L_+^g T_+^h &= T_+^{gh} L_+^g, & L_+^g L_-^h &= T_-^{hg^{-1}} L_+^g \\ L_-^g T_+^h &= T_+^{hg^{-1}} L_-^g, & L_-^g T_-^h &= T_-^{gh} L_-^g \end{aligned}$$

Hilbert space:



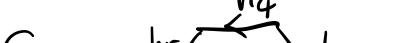
$|G|$ -dim vector space on
each link of a 2D lattice.

$$\overrightarrow{\bullet} = \overleftarrow{\bullet}, \quad g \in G.$$

QDM : $H = - \sum_s A_s - \sum_p B_p$

$$\backslash \quad A_s^g := \prod_{\substack{\text{edge } l \\ \text{of } s}} L_{\pm}^g, \quad A_s := \frac{1}{|G|} \sum_{g \in G} A_s^g$$

↳ gauge transf.

B_p

 B_p^h

$$B_p^h = \sum_{h_1 \dots h_6 = h} \prod_{m=1}^6 T_{\pm}^{h_m}, \quad B_p := B_p^{h=1}$$

↪ zero flux condition for p.

relation $\left\{ \begin{array}{l} A_s^2 = A_s \\ B_p^2 = B_p \end{array} \right\}$ projectors

$[A_s, A_{s'}] = 0$

$[B_p, B_{p'}] = 0$

$[A_s, B_p] = 0$ ↘ proof

A_s

$$\begin{array}{ccc} g_2 \downarrow & \xrightarrow{\quad \phi \quad} & g_2 g_1 \\ s & \xrightarrow{\quad g_1 \quad} & g_1 \end{array} = \begin{array}{ccc} g_2 g_1 \downarrow & \xrightarrow{\quad \phi \quad} & (g_2 g_1) \cdot (g_1 g_3) = g_2 g_1 \\ s & \xrightarrow{\quad g_1 \quad} & g_1 \end{array}$$

Ground state : $A_s |\psi\rangle = B_p |\psi\rangle = |\psi\rangle$, $\forall s, p$
↓ gauge transf. ↓ flat connection condition

$\Rightarrow |\psi\rangle$: flat connections up to gauge equivalence.

T.C. $|\psi\rangle = | \bigcirc \rangle + | \bigodot \rangle + \dots$
 $g = e^{iA} \in U(1)$

flat : $\oint_C A_\mu dx^\mu = 0$

$g_{ij} = e^{iA_{ij}}$ ↗ continuous
↘ discrete

$\prod g_1 \dots g_N = 1 \Leftrightarrow B_p^{h=1}$

$g = e^{iA}$
 $g^{-1} = e^{-iA}$
 r
 $g = e^{iA}$
 $g = e^{-iA}$

Gauss law

$\nabla \cdot \vec{A} = \rho$

$\parallel$

$A(\vec{r} + \hat{x}) - A(\vec{r})$

$+ A(\vec{r} + \hat{y}) - A(\vec{r})$

$\downarrow$

$g_{\vec{r}+\hat{x}} g_{\vec{r}}^{-1} g_{\vec{r}+\hat{y}} g_{\vec{r}}^{-1}$

Important notice:

The courses of week 3 (on 2021-09-27 and 2021-09-29) will be cancelled. And the ending date of the course will be postponed to week 13 of the fall semester.

From 2021-10-04 on, we will use another Tencent Meeting Room [9952830954](#) (password 654321) of the university account, such that there are more storage space for the recordings.

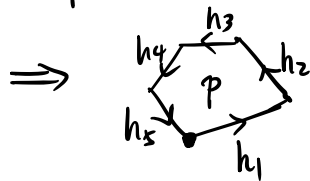
Recall. QDM.

$$H = - \sum_s A_s - \sum_p B_p$$

$$[A_s, B_p] = 0$$

$$\textcircled{1} B_p |\psi\rangle = |\psi\rangle, \quad \forall p$$

$$\xrightarrow{h} = \xleftarrow{h^{-1}}$$



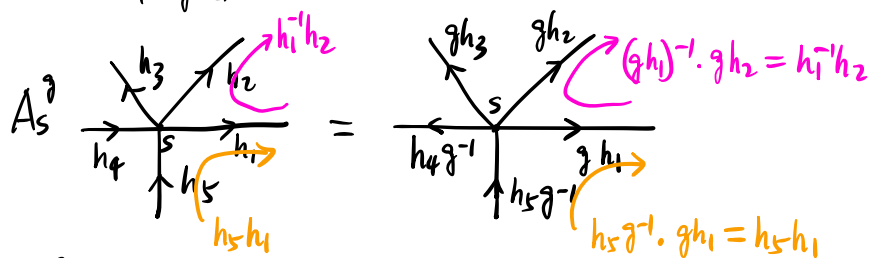
$$h_1 h_2 \cdots h_5 = 1 \in G.$$

(flat connection condition)
(zero flux condition)

$$\textcircled{2} A_s |\psi\rangle = |\psi\rangle, \quad \forall s$$

$$A_s := \frac{1}{|G|} \sum_{g \in G} A_s^g$$

A_s^g : gauge transformation.



A_s^g fluctuate $\{h_{ij}\}$ within the subspace $B_p=1, \forall p$.

$\Rightarrow G_s |\psi\rangle$ is an equal weight superposition of all flat connections that are gauge equivalent.

continuous: $U(1)$ gauge theory

$$A_\mu(x) \rightarrow A'_\mu = A_\mu - \partial_\mu \chi$$

$$\oint_C A'_\mu dx^\mu = \oint_C (A_\mu - \partial_\mu \chi) dx^\mu = \oint_C A_\mu dx^\mu = \oint$$

is gauge inv.

discrete: $i \xrightarrow{g_{ij}} j$

$$g_{ij} = e^{i \int_{\text{site } i}^{\text{site } j} A_\mu dx^\mu}$$

→ gauge field

$$g_i = e^{i \chi(r_i)}$$

→ gauge transf. para.

gauge transf. ↓

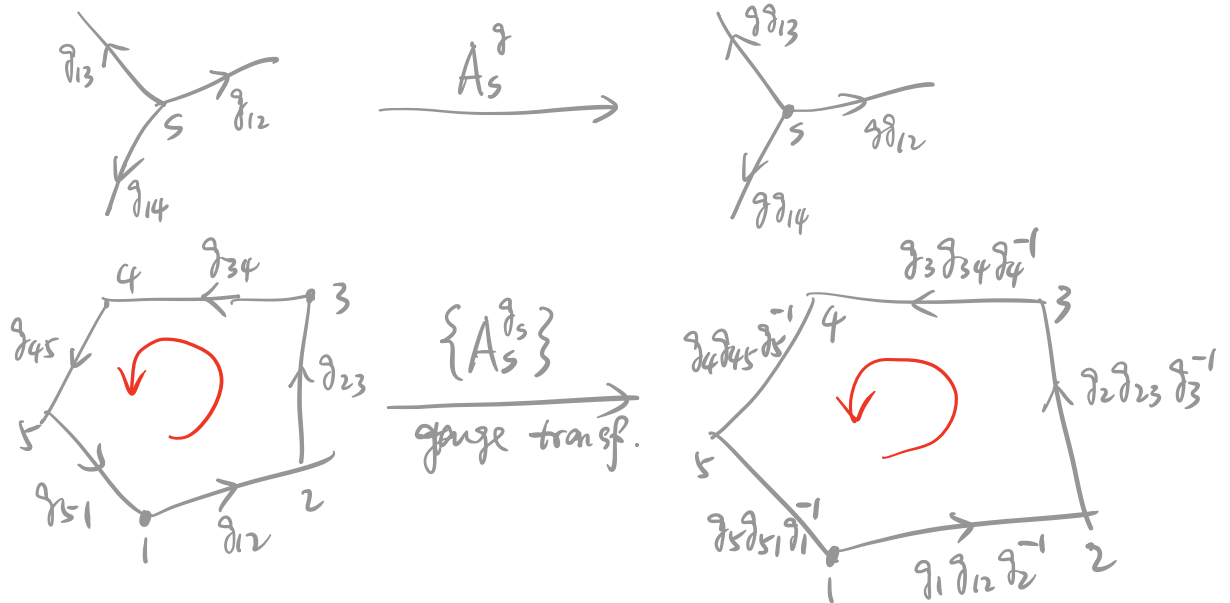
$$g'_{ij} = e^{i \int_i^j (A_\mu - \partial_\mu \chi) dx^\mu} = e^{i \int_i^j A_\mu dx^\mu - \chi(r_j) + \chi(r_i)}$$

$$= e^{i \chi(r_i)} e^{i \int_i^j A_\mu dx^\mu} e^{-i \chi(r_j)}$$

$$= g_i \cdot g_{ij} \cdot g_j^{-1}$$

$i \xrightarrow{g'_{ij}} j$
 $g'_{ij} = g_i g_{ij} g_j^{-1}$

eg:



$$\oint = g_{12} g_{23} \dots g_{51}$$

$$= e^{i \int_1^1 A_\mu dx^\mu}$$

$$\oint' = g_1 g_{12} g_2^{-1} \cdot g_2 g_{23} g_3^{-1} \cdot \dots \cdot g_5 g_{51} g_1^{-1}$$

$$= g_1 (g_{12} \dots g_{51}) g_1^{-1}$$

$$= g_1 \oint g_1^{-1}$$

$$\text{Tr } \oint' = \text{Tr } \oint$$

2.2. Anyon excitations and ribbon operators

$$H = - \sum_s \underbrace{\frac{1}{|G|} \sum_{g \in G} A_s^g}_{\text{proj. to gauge inv. subspace}} - \sum_p \underbrace{B_p^{h=1}}_{\text{proj. to } h=1 \text{ zero flux}}$$

Excitations: ① non gauge inv.: $\frac{1}{|G|} \sum_g A_s^g \neq 1$
 ② non zero flux: $B_p^{h \neq 1}$

To understand the excitations of QDM, we first try to understand the algebra of A_s^g, B_p^h for 

$$\begin{cases} A^g A^{g'} = A^{gg'} \\ (A^g)^+ = A^{g^{-1}} \\ B^h B^{h'} = \delta_{hh'} B^h \\ (B^h)^+ = B^h \\ A^g B^h = B^{ghg^{-1}} A^g \end{cases}$$

note: (A_s^g, B_p^h) for different (s, p) commute with each other and are isomorphic.

Def $D_{(h,g)} = h \underset{g}{\llcorner} := B_h \cdot A_g$

$$D_{(h_1, g_1)} \cdot D_{(h_2, g_2)} = \delta_{h_1, g_1 h_2 g_1^{-1}} D_{(h_1, g_1 h_2)}$$

The algebra generated by $D_{(h,g)}$ is called the Drinfeld quantum double $D(G)$ of G .

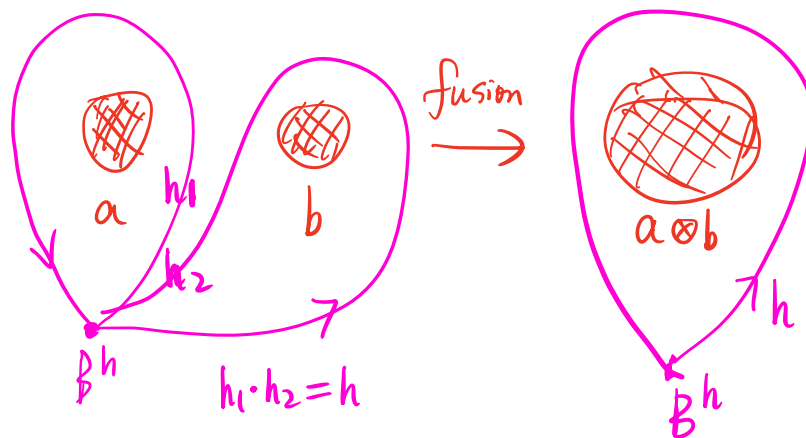
① Co-algebra structure $\Delta: D(G) \rightarrow D(G) \times D(G)$ → how to act on two anyons by one $D(G)$ element

$\Delta(A^g) = A^g \otimes A^g$ → act on anyon a

$\Delta(B^h) = \sum_{h_1 h_2 = h} B^{h_1} \otimes B^{h_2}$

$\Rightarrow \text{Rep } D(G)$ is a tensor/monoidal category.

$$a \otimes b \rightarrow c$$



② $D(G)$ is quasi-triangular: $R = \sum_{g \in G} B^g \otimes A^g$
braiding

$\Rightarrow \text{Rep } D(G)$ is a braided tensor category.

$D(G)$ is a quasi-triangular Hopf algebra.

Excitation space supports a representation of $D(G)$.

Irreps of quantum double $D(G)$
 $\longleftrightarrow$ anyon excitation types

described by braided fusion cat.

Irreps of $D(G)$.

Let $u \in G$, $C = \{g u g^{-1} \mid g \in G\}$ the conjugacy class,
 $E = \{g \in G \mid g u = u g\}$ the centralizer.

Claim. There is one irreducible rep (C, χ)
for each conjugacy class C and each
irrep χ of the centralizer group E .

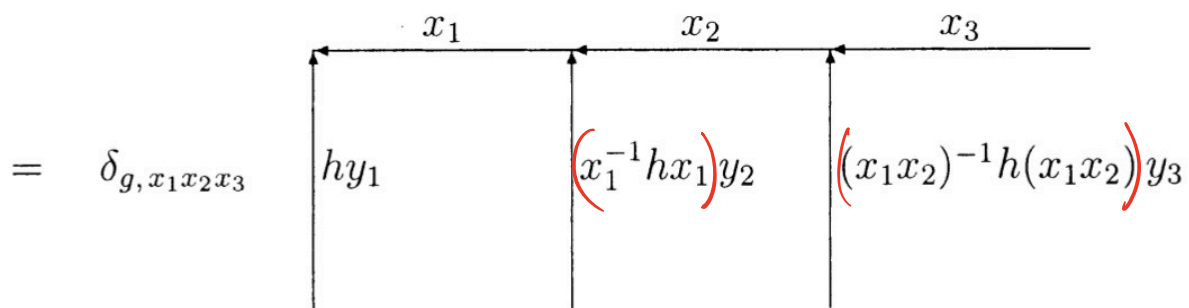
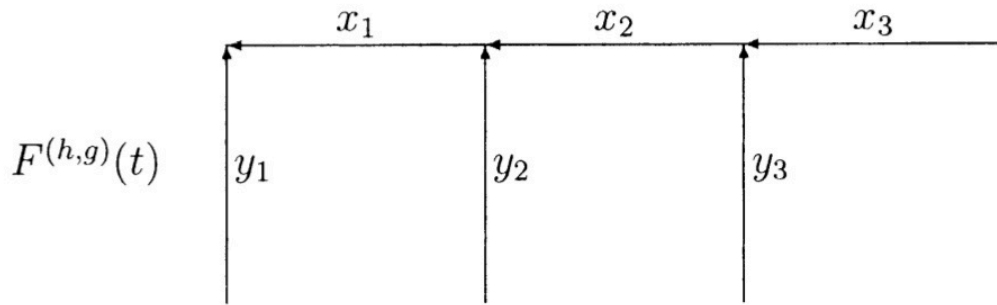
flux $\uparrow$
change $\rightarrow$

Ribbon operators



Create a pair of anyonic excitations at the ends of the ribbon operators.

↳ finite width (framing)

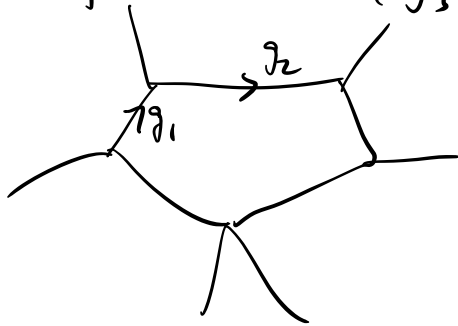


$$\begin{aligned} [F^{(h,g)}(t), B_p] &= 0 \\ [F^{(h,g)}(t), A_s] &= 0 \end{aligned} \quad \text{for } s, p \neq \partial t$$

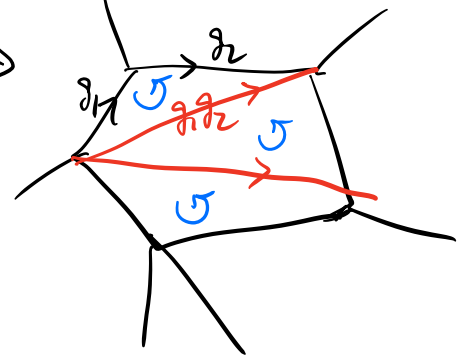
F gives us fusion, braiding of anyons.
(see Kitaev's original paper)

2.3. Spacetime Path Integral Picture

$\forall$ 2D lattice
flat connection $\{g_{ij}\}$



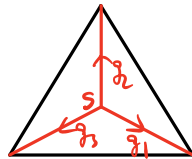
2D triangulation
flat connection



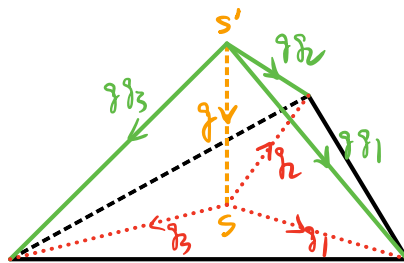
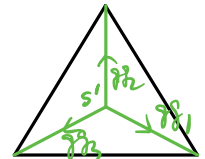
B_p operator: defined for $\forall$ triangle. flat connection
for all spacial triangles
 A_s^g operator:

$$\partial(\text{spacetime}) = \text{space}(t_2) - \text{space}(t_1)$$

space :
spacetime :

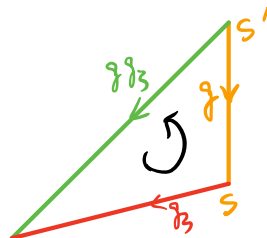


A_s^g

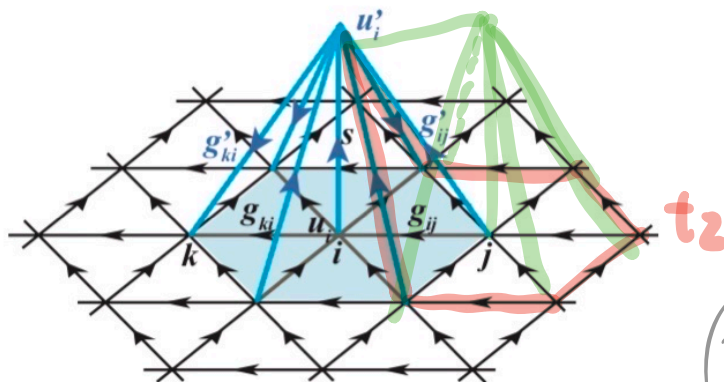


t

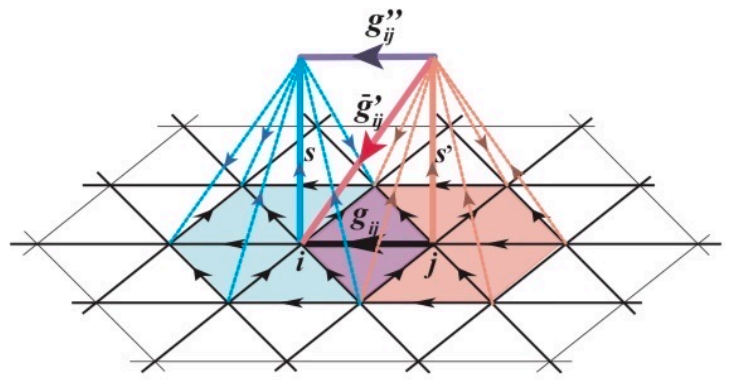
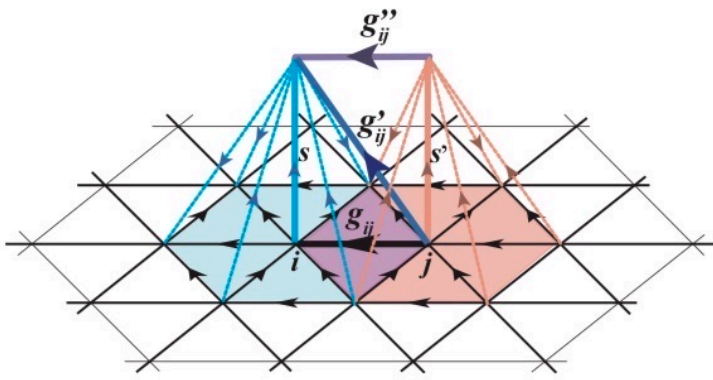
new link g along t direction



flat connection for all
spacetime triangles.



(figs from Mesaros, Ran)
arXiv:1212.0835



$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ t_3 & t_2 & t_1 \end{matrix} A_j^{g'} A_i^g$$

=
exercise

$$\begin{matrix} \uparrow & \uparrow & \uparrow \\ t_3 & t_2 & t_1 \end{matrix} A_i^g A_j^{g'}$$

• Partition function

discrete: $Z_{QDM} = \frac{1}{N} \sum_{\{g_{ij}\}} \prod_{\langle ij \rangle} \underbrace{\delta_{g_{ij} g_{jk} g_{ik}}}_{0 \text{ or } 1}$

flat connection for triangle $\langle ij \rangle$



continuum: $Z_{QDM}(M_3) = \frac{1}{|G|} \sum_{\gamma \in \text{Hom}(\pi_1(M_3), G)} 1 \rightarrow e^{-E(\{g_{ij}\})}$

where $\gamma: \pi_1(M_3) \rightarrow G = \pi_1(BG)$

$E(\{g_{ij}\}) = \begin{cases} 0, & \text{flat} \\ \infty, & \text{non-flat} \end{cases}$

loop $c \mapsto \underbrace{\oint_c A}_{\text{flux}} = e^{i \oint_c A} = \prod_{\langle ij \rangle \in c} g_{ij}$

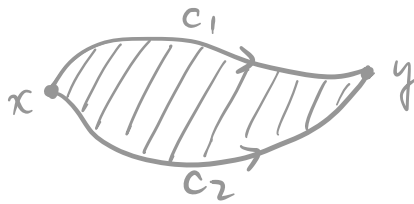
$1 \in G \quad \oint = 1 \in G$

$\oint = 1 \in G$

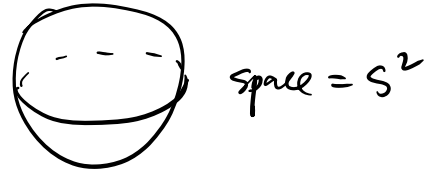
$\oint(c)$

note: loops $c_1 \sim_{\text{homotopy}} c_2$

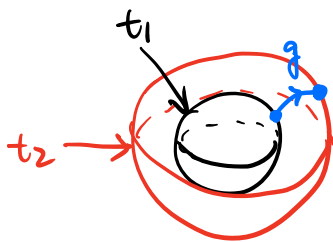
$\Rightarrow \oint(c_1) = \oint(c_2)$ for flat connections.



Example: (1) $M_3 = S^2 \times S^1$



$\pi_1(S^2) = 0 \Rightarrow$ All flat connections are gauge equivalent.



$$\gamma: \pi_1(S^2 \times S^1) = \mathbb{Z} \rightarrow G$$

$$|\{\text{flat conn.}\}| = |G|$$

$$\Rightarrow Z_{\text{QDM}}(S^2 \times S^1) = \frac{1}{|G|} |G| = 1$$

= ground state degeneracy (S^2)
GSD

(1) $M_3 = T^2 \times S^1 = T^3$, $G = \mathbb{Z}_2 \rightarrow$ toric code model.

$$\gamma: \pi_1(T^3) = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \rightarrow G = \mathbb{Z}_2$$

$$|\{\text{flat conn.}\}| = 2^3 = 8$$

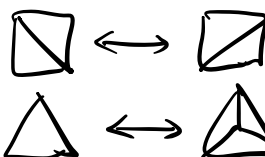
$$Z_{\text{QDM}}(T^3) = \frac{1}{|G|} |\{\text{flat conn.}\}| = \frac{1}{2} \times 8 = 4$$

$$= \text{GSD}(T^2)$$

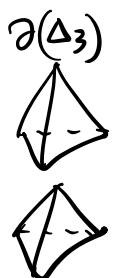
• Retriangulation invariance of Z_{QDM} .

Pachner moves.

eg: 2D
2-2 move
1-3 move



glue
glue



Any two triangulations of a piecewise linear manifold can be related by a finite sequence of Pachner moves.

(1) Path integral $(1+1D)$

$$Z \left(\text{diagram 1} \right) = Z \left(\text{diagram 2} \right)$$

$$Z \left(\text{diagram 3} \right) = Z \left(\text{diagram 4} \right)$$

flat connection $\xrightarrow[\text{local patch change}]{\text{Pachner move}}$ flat connection

$$Z_{\text{ADM}}(M, T; G) = Z_{\text{ADM}}(M, T'; G)$$

$\uparrow$ spacetime manifold $\uparrow$ triangulation

(2) wave function

$$(2+1)D : |\Psi_{\text{GS}}\rangle = \sum_{\substack{\text{flat conn.} \\ \{g_{ij}\}}} \Psi(\{g_{ij}\}) |\{g_{ij}\}\rangle$$

$$\Psi(\{g_{ij}\}) = \Psi(\{g'_{ij}\}) \text{ if } \{g_{ij}\} \stackrel{\text{gauge equiv.}}{\sim} \{g'_{ij}\}$$

$\uparrow$
equal amplitude superposition of gauge equiv. flat connections.

generalization from one fix lattice to arbitrary lattice:

$$|\Psi_{\text{GS}}\rangle = \sum_{\substack{\text{lattice } T \\ (\text{triangulation})}} \frac{1}{N_T} \sum_{\text{flat conn. } \{g_{ij}\}} \Psi(\{g_{ij}\}) |\{g_{ij}\}\rangle$$

$$\psi(\square) = 1 \times \psi(\square)$$

$\downarrow \text{QDM}$

$$U_3(g, h, k)$$

$\downarrow \text{TQDM}$

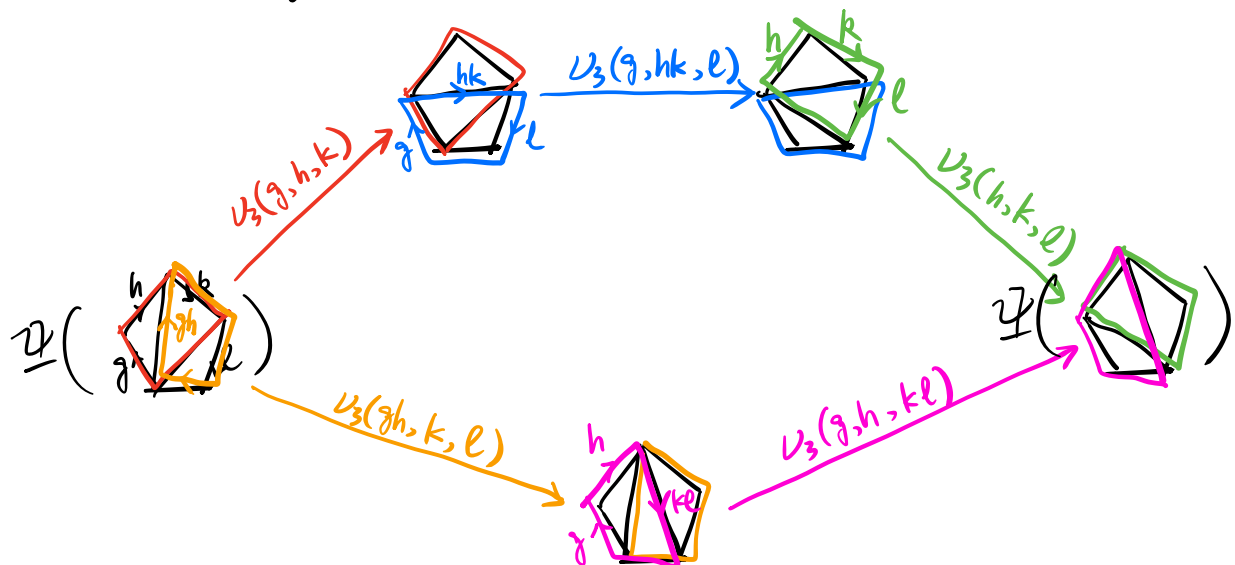
2.4. Dijkgraaf-Witten gauge theory as twisted QDM.
(1990)

- Introduce d-coycle for wavefunction retriangulation

$$\psi \left(\begin{array}{c} \downarrow \\ \text{square with } g, h, k \end{array} \right) = \underbrace{\psi_3(g, h, k)}_{\text{triangle}} \quad \psi \left(\begin{array}{c} \downarrow \\ \text{square with } g, h, k \end{array} \right)$$

function $U_3: G^3 \rightarrow U(1)$

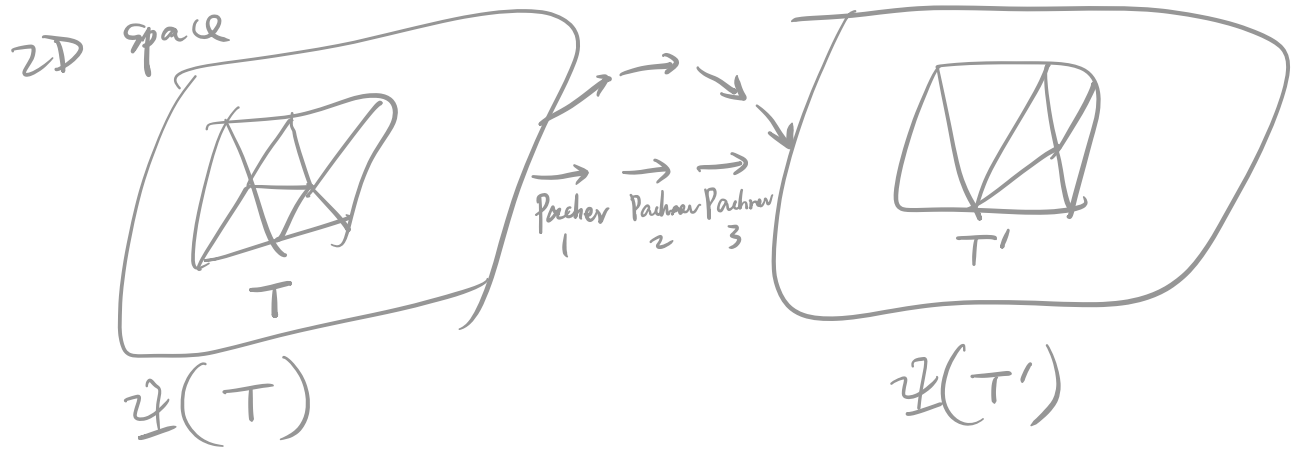
Pentagon equation as consistency condition :



$$\nu_3(g, h, k) \nu_3(g, hk, l) \nu_3(h, k, l) = \nu_3(gh, k, l) \nu_3(g, h, kl)$$

$$\Leftrightarrow (d\nu_3)(g, h, k, l) := \frac{\nu_3(h, k, l) \nu_3(g, h, k, l) \nu_3(g, h, k)}{\nu_3(g, h, k, l) \nu_3(g, h, k, l)} = 1$$

$$\Leftrightarrow \nu_3 \in Z^3(G, U(1)) \text{ is a 3-cocycle.}$$



- Spacetime Path integral on the lattice

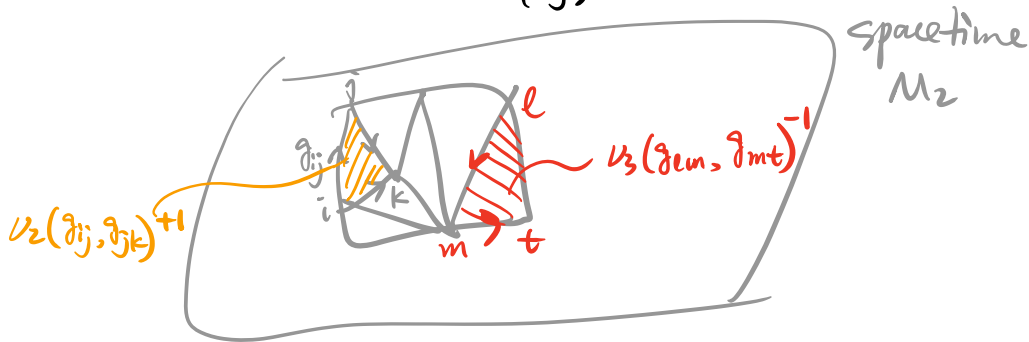
1+1D:

1-space: $\psi(\text{---} \xrightarrow{gh} \text{---}) \xrightarrow[\text{move}]{1-2} \nu_2(g, h) \psi(\text{---} \xrightarrow{g} \text{---} \xrightarrow{h} \text{---})$

2-spacetime:

$\begin{array}{c} g \nearrow \\ \triangle \\ g \searrow \\ h \\ \text{---} \end{array} = \nu_2(g, h)$

$$Z_{\text{TQFT}}(\mathcal{M}_2) := \frac{1}{N} \sum_{\text{flat conn. } \{g_{ij}\}} \prod_{\substack{\langle ijk \rangle \\ \in T}} \nu_2(g_{ij}, g_{jk}) \underbrace{s(ijk)}_{\substack{= \pm 1 \text{ for orientations}}}$$



retriangulation of $\mathcal{M}_2$:

$$\underbrace{\begin{array}{c} \xrightarrow{h} \\ \begin{array}{|c|} \hline \begin{array}{c} g \nearrow \quad \searrow k \\ \hline \end{array} \\ \hline \end{array}}_{\nu_2(g, h) \nu_2(g, h, k)} = \underbrace{\begin{array}{c} \xrightarrow{h} \\ \begin{array}{|c|} \hline \begin{array}{c} \nearrow h \quad \searrow k \\ \hline \end{array} \\ \hline \end{array}}_{\nu_2(g, h, k) \nu_2(h, k)}$$

$$\Leftrightarrow (d\nu_2)(g, h, k) = 1$$

$$\Leftrightarrow Z(\partial \Delta_3) = Z\left(\begin{array}{c} \text{triangle} \\ \text{with edges } g, h, k \end{array}\right) = Z(S^2) = 1$$

2+1 D:

$$2\text{-space: } \psi\left(\begin{array}{c} \square \\ \diagup \end{array}\right) = \nu_3(g, h, k) \quad \psi(\square)$$

$$3\text{-spacetime: } \begin{array}{c} \text{tetrahedron} \\ \text{with edges } g, h, k \end{array} = \nu_3(g, h, k)$$

$$Z_{\text{TQDM}} := \frac{1}{N} \sum_{\substack{\text{flat} \\ \text{conn.} \\ \{g_{ij}\}}} \prod_{\substack{\langle ijkl \rangle \\ \in T}} \nu_3(g_{ij}, g_{jk}, g_{kl})^{S(ijkl)}$$

retriangulation of M_3 :

$$Z\left(\begin{array}{c} \text{tetrahedron} \\ \text{with edges } g, h, k \end{array}\right) \underset{\nu_3 \cdot \nu_3}{=} Z\left(\begin{array}{c} \text{tetrahedron} \\ \text{with edges } g, h, k, l \end{array}\right) \underset{\nu_3 \nu_3 \nu_3}{=}$$

$$\Leftrightarrow d\nu_3 = 1$$

$$\Leftrightarrow Z(\partial \Delta_4) = Z\left(\begin{array}{c} \text{tetrahedron} \\ \text{with edges } g, h, k, l \end{array}\right) = Z(S^3) = 1$$

Generalize to dD:

$$d\text{-spacetime: } d\text{-simplex } \Delta_d \mapsto \nu_d(g_1, \dots, g_d)$$

$$\text{consistency condition: } \nu_d(\partial \Delta_{d+1}) = \nu_d(S^d) = 1$$

partition function:

$$Z_{\text{TQDM}}(M_d, T; G) = \frac{1}{N} \sum_{\substack{\text{flat conn.} \\ \{g_{ij}\}}} \prod_{\Delta_d \in T} \nu_d(\Delta_d)^{S(\Delta_d)}$$

$$Z_{\text{TQDM}}(M_d, T; G) = Z_{\text{TQDM}}(M_d, T'; G)$$

- Spacetime path integral in the continuum:
(DW in 1990)

$$Z_{QDM} = \frac{1}{|G|} \sum_{\gamma \in \text{Hom}(\pi_1(M_d), G)} 1$$

↓ twist by $\nu_d \in Z^d(G, U(1))$

$$Z_{TQDM} = Z_{DW} = \frac{1}{|G|} \sum_{\gamma \in \text{Hom}(\pi_1(M_d), G)} \langle \gamma^* \nu_d, [M] \rangle$$

$$\gamma^* \nu_d : \pi_1(M_d) \xrightarrow{\gamma} G \xrightarrow{\nu_d} U(1)$$

$$\langle \gamma^* \nu_d, [M] \rangle \sim e^{i \int_{M_d} \tilde{\nu}_d} \sim \prod_{\Delta_d \in T} \nu_d(\Delta_d)^{s(\Delta_d)}$$

↑
additive $\nu_d = e^{i \tilde{\nu}_d}$